

2. For the Web application automate the continuous integration and deployment process using Jenkins. ⌚ Write the code in Visual Studio and push it to your GITHUB repository. ⌚ Modify the code and commit the updates back in GITHUB. ⌚ Using Jenkins configure the job as pipeline and create the stages for the Continuous Integration process. ⌚ Checkout GITHUB ⌚ Build using Maven ⌚ Create Docker image ⌚ Push to Docker Hub ⌚ Create Cluster in Kubernetes and Deploy

To display the payslip of the employee

The screenshot shows a GitHub repository named 'web-ci-cd' by user 'shalini-s2023'. The repository files include 'Batch1 and Batch1', 'Dockerfile', 'README.md', 'deployment.yaml', 'index.html', and 'jenkinsfile'. The 'index.html' file is open in the Visual Studio code editor, displaying an HTML document with a title 'payslip' and a table showing employee details for 'shalini'.

year	amount
2020	60000
2021	60000
2022	62000

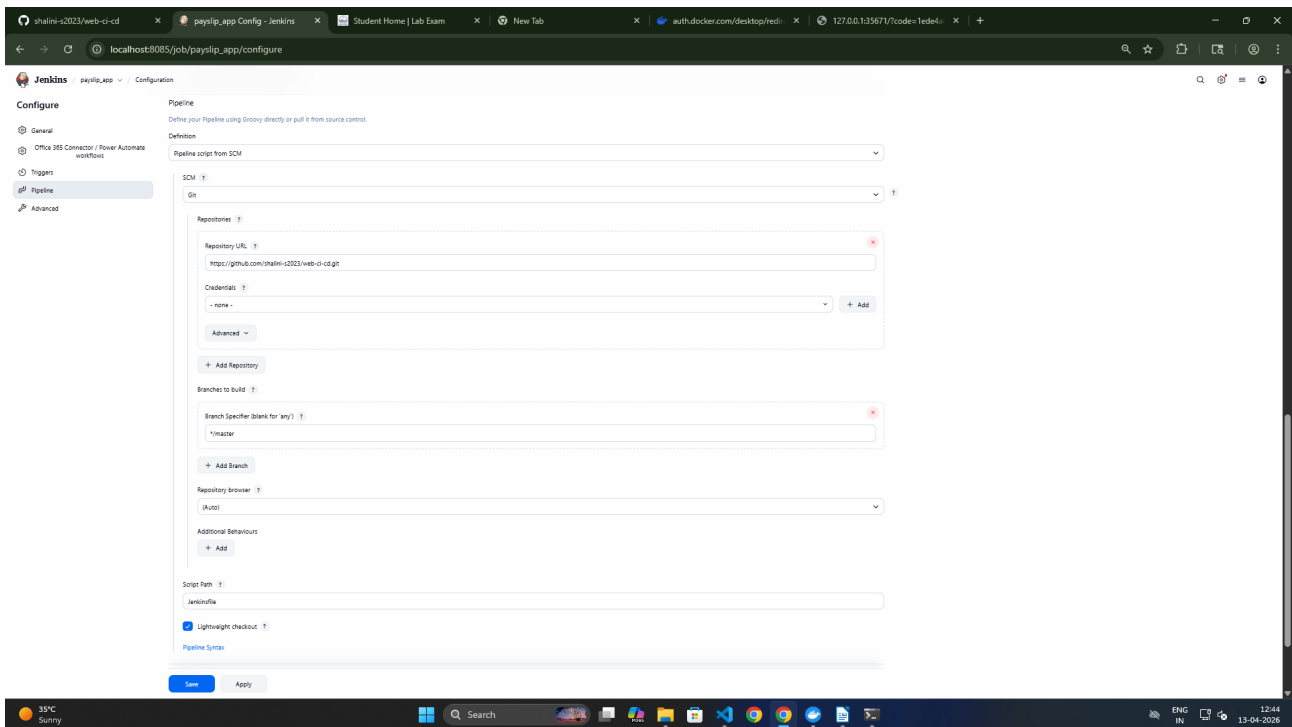
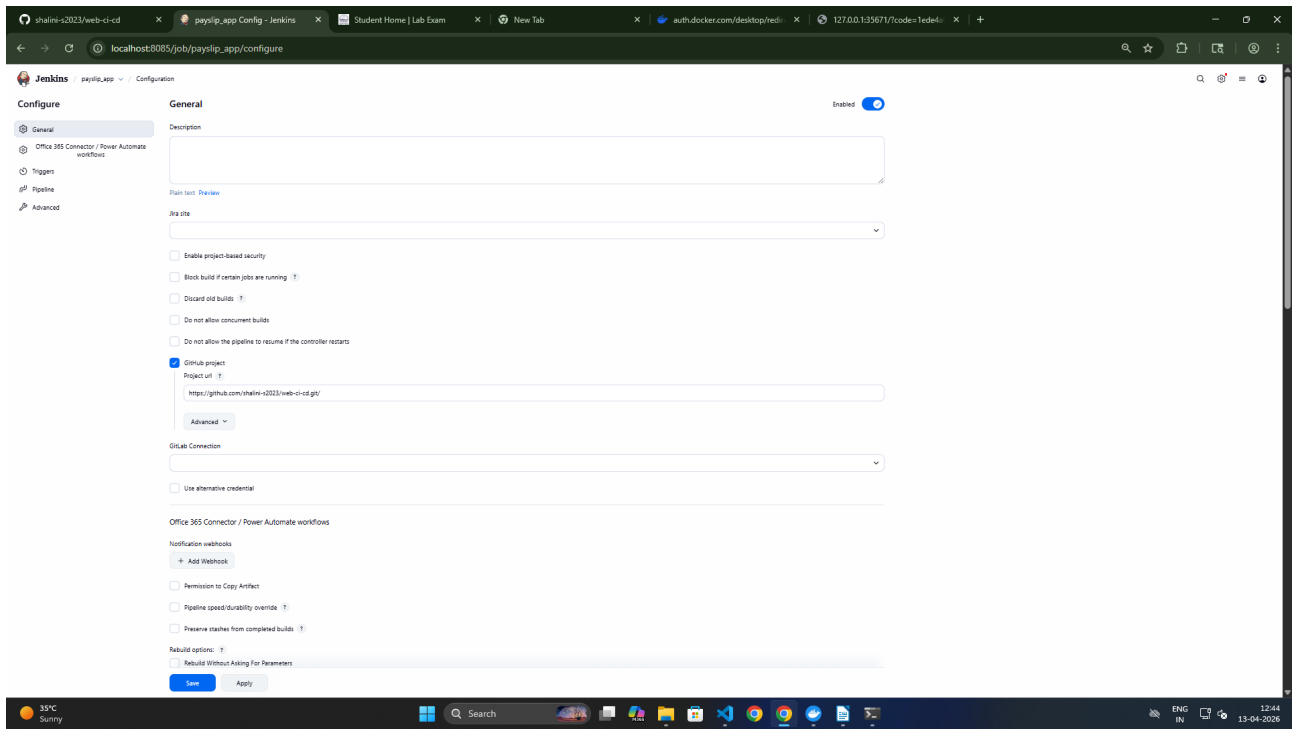
The Visual Studio code editor also shows a terminal window with the following output:

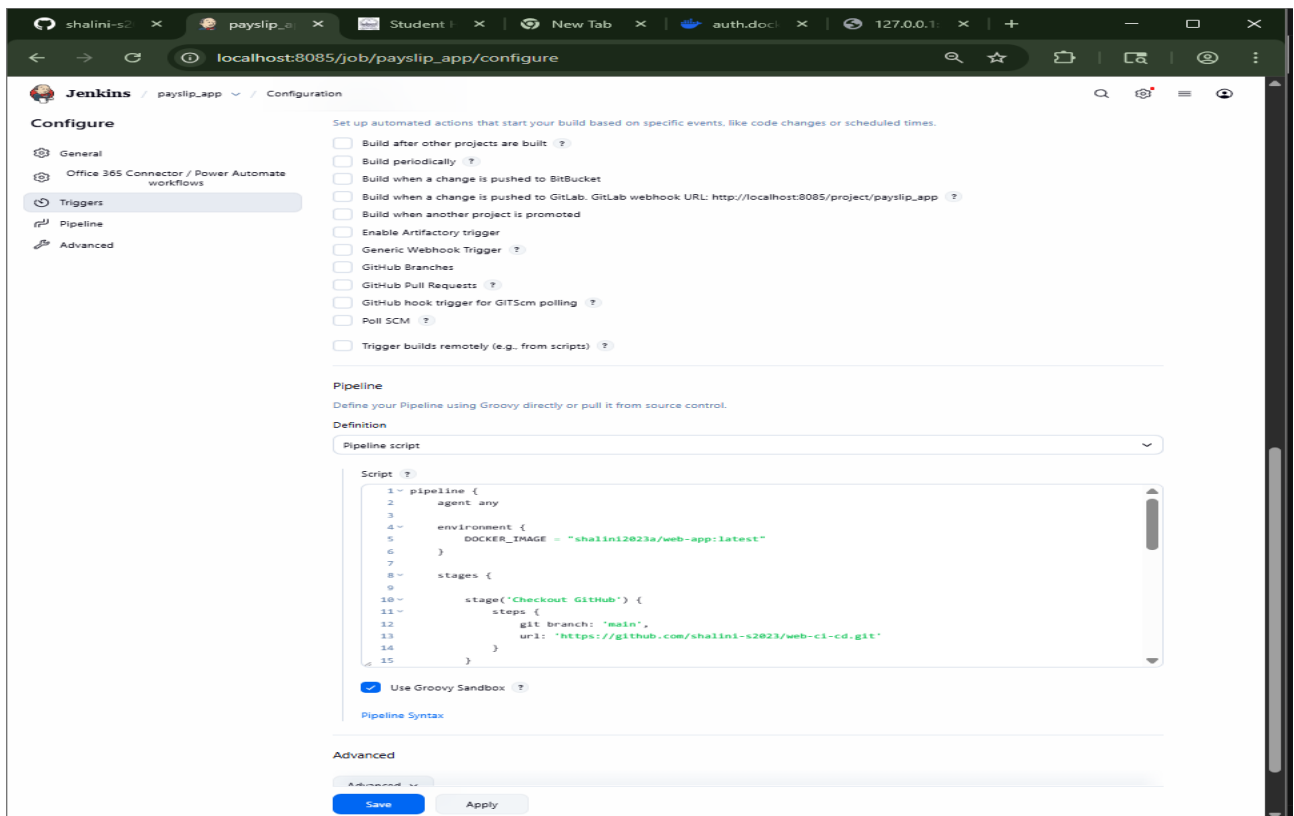
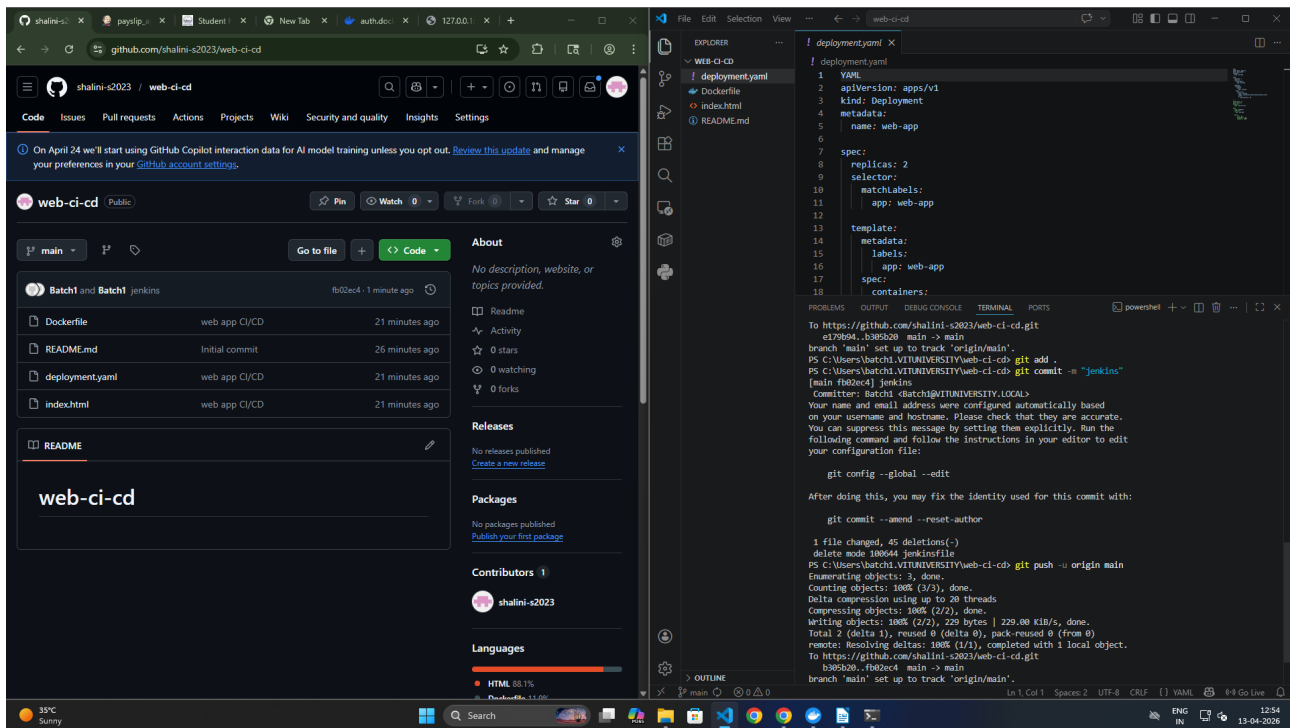
```
git commit --amend --reset-author
4 files changed, 61 insertions(+)
ch -m main
PS C:\Users\batch1.VITUNIVERSITY\web-ci-cd> git remote add origin https://github.com/shalini-s2023/web-ci-cd.git
error: remote origin already exists.
PS C:\Users\batch1.VITUNIVERSITY\web-ci-cd> git push -u origin main
info: please complete authentication in your browser...
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 20 threads
Compressing objects: 100% (5/5), done.
Writing objects: 100% (6/6), 953 bytes | 953.00 KiB/s, done.
Total 6 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/shalini-s2023/web-ci-cd.git
d2b38a1..e179b94 main -> main
branch 'main' set up to track 'origin/main'.
```

The screenshot shows the Jenkins 'New Item' configuration page. The 'Enter an item name' field is filled with 'payslip_scd'. The 'Select an item type' dropdown is set to 'Pipeline'. The 'Pipeline' option is selected, and its description is visible: 'Build, test, and deploy using pipelines. Supports stages, parallel work, and running on multiple agents.'

Other available item types include:

- Freestyle project
- Maven project
- External job
- Multi-configuration project
- MultiJob Project
- hvy project
- Folder
- Multibranch Pipeline
- Multibranch Pipeline with defaults





shalini-s2 ×payslip_app ×Student ×New Tab ×auth.doc1 ×127.0.0.1 ×+

localhost:8085/job/payslip_app/

Jenkins / payslip_app

Status

Changes

Build Now

Configure

Delete Pipeline

Full Stage View

GitHub

Favorite

Job Config History

Open Blue Ocean

Stages

Rename

Pipeline Syntax

Set Next Build Number

Builds

Filter

Today

#5 1:07 PM

#4 1:01 PM

#3 12:58 PM

#2 12:51 PM

#1 12:50 PM

payslip_app

Add description

Stage View

Average stage times:

	Checkout GitHub	Build Docker Image	Push to DockerHub	Deploy to Kubernetes
#5 Apr 13 13:07 No Changes	1s	2s	9s	461ms failed
#4 Apr 13 13:01 No Changes	2s	12s	18s	660ms failed
#3 Apr 13 12:58 No Changes				
#2 Apr 13 12:51 No Changes				
#1 Apr 13 12:50 No Changes				

Permalinks

Last build (#4), 5 min 18 sec ago

Last failed build (#4), 5 min 18 sec ago

Last unsuccessful build (#4), 5 min 18 sec ago

Last completed build (#4), 5 min 18 sec ago

Help us localize this page

REST API Jenkins 2.528.3

File Edit Selection View ...

web-ci-cd

EXPLORER

WEB-CI-CD

deployment.yaml

Dockerfile

index.html

README.md

deployment.yaml

index.html

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43

! deployment.yaml

spec:

apiVersion: v1

kind: Service

metadata:

name: web-service

spec:

type: NodePort

selector:

app: web-app

ports:

- port: 80

targetPort: 80

nodePort: 30007

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

powershell

+

-

...

git config --global --edit

After doing this, you may fix the identity used for this commit with:

git commit --amend --reset-author

1 file changed, 29 insertions(+), 8 deletions(-)

PS C:\Users\batch1.VITUNIVERSITY\web-ci-cd> git push -u origin main

Enumerating objects: 5, done.

Counting objects: 100% (5/5), done.

Delta compression using up to 20 threads

Compressing objects: 100% (3/3), done.

Writing objects: 100% (3/3), 507 bytes | 507.00 KiB/s, done.

Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)

remote: Resolving deltas: 100% (1/1), completed with 1 local object.

To https://github.com/shalini-s2023/web-ci-cd.git

cc475d9..cd4e020 main -> main

branch 'main' set up to track 'origin/main'.

PS C:\Users\batch1.VITUNIVERSITY\web-ci-cd> kubectl get svc

NAME TYPE CLUSTER-IP EXTERNAL-IP PORT(S) AGE

kubernetes ClusterIP 10.96.0.1 <none> 443/TCP 3d21h

PS C:\Users\batch1.VITUNIVERSITY\web-ci-cd> kubectl get pods

> OUTLINE

main

0

0

Batch1 (now)

Ln 42, Col 1

Spaces: 2

UTF-8

CRLF

{ }

YAML

Go Live

13:32

13-04-2026

|

git link : <https://github.com/shalini-s2023/web-ci-cd.git>